Surveying Engineering

Introduction

Instructor

Heeseung Shon, Ph.D.

Course Objectives

- **This course aims to develop students' knowledge and skills in:**

    A. Conducting geospatial measurements for common civil engineering projects;

    B. Understanding the types of surveying instruments, measurement techniques, and field procedures used to collect geospatial data;

    C. Collecting, processing, analyzing, and plotting geospatial data required to produce engineering-scale maps;

    D. Understanding spatial reference systems, coordinate systems, and positioning frameworks; and

    E. Formulating, solving, and clearly presenting solutions to surveying, positioning, and location-related problems.

Textbook and References

- Lecture Note
- Surveying: Principles and Applications, B. Kavanagh, Pearson

Class Contents

Basics of Surveying

Leveling

Random Error

Distance Measurement

Angle Measurement and Traverse

Topographic Surveying

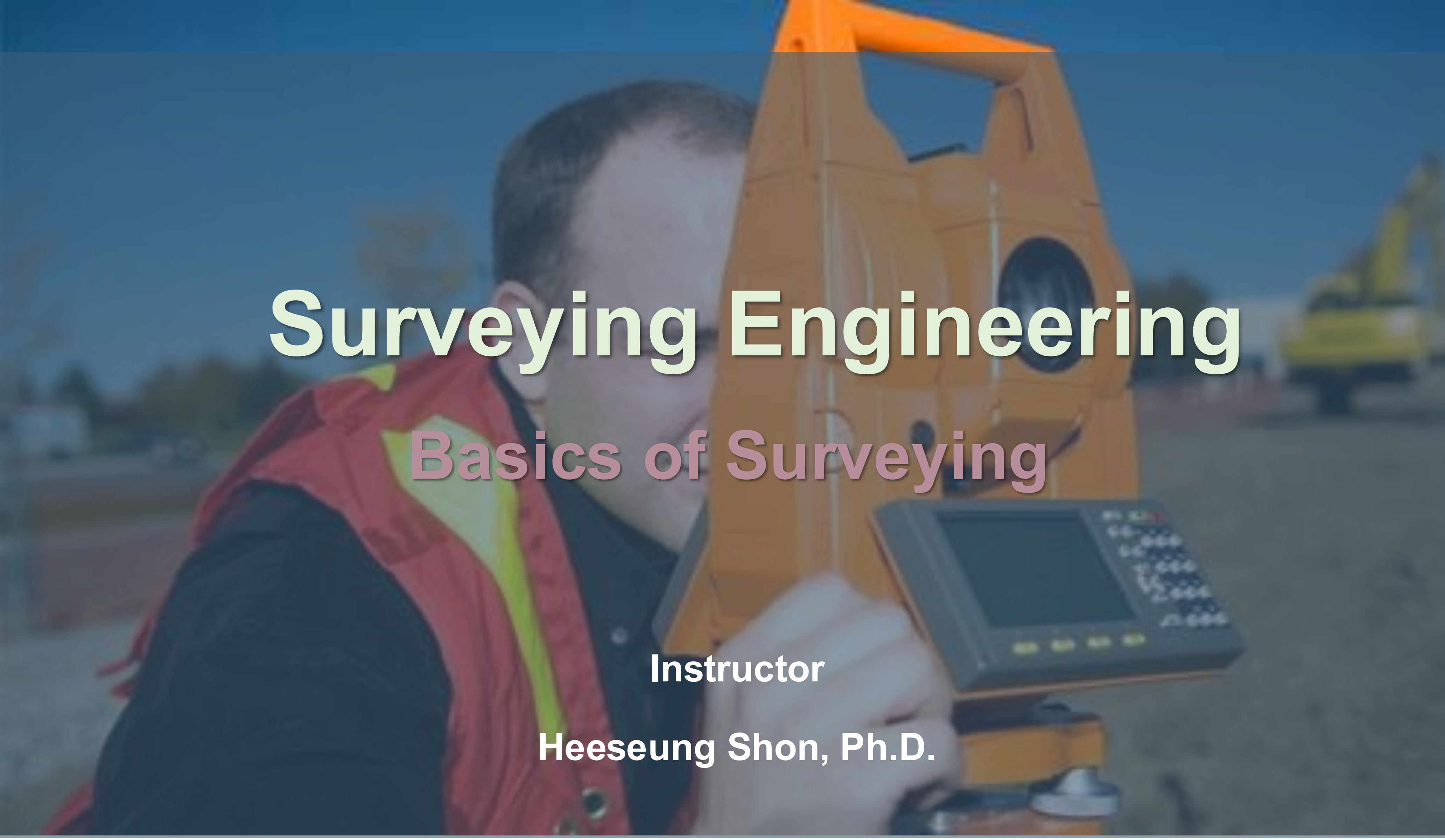
Earth Shape and Map Projection

Coordinate Systems

Global Positioning System

Remote Sensing

Geographic Information System (GIS)



Surveying Engineering

Basics of Surveying

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Definition of Survey

❑ Definition (Wolf, 1994)

- Science and technology of determining the relative positions of points above, on, or beneath the earth's surface, or of establishing points.
- Discipline which encompasses all methods for measuring, processing, and disseminating information about the physical earth and our environment.

❑ Target, Components, and Methods

- Target: the space for human being (e.g., earth, water, underground, sea)
- Components: length, angle, time, mass and gravity
- Methods: gravity, earth magnetic, electromagnetic wave, temperature, light

Definition of Survey

❑ Role of Surveying

- ❖ Determining the shape and dimensions of the Earth
- ❖ Establishing and refining spatial reference frameworks
- ❖ Locating existing natural and constructed features
- ❖ Setting out designed positions of engineering structures
- ❖ Producing maps, plans, charts, and digital survey records

Role of Surveying

Determining the shape and dimensions of the Earth

- **Early Ideas of Earth**

  - Earth stood still - otherwise, we would feel ourselves moving
  - Must be flat - otherwise we would fall off
  - Immense - stretching into unknown oceans and lands
  - World center of the universe - looking up, one saw the moon and stars revolving about the earth
  - Each society thought itself as the center of universe

Role of Surveying

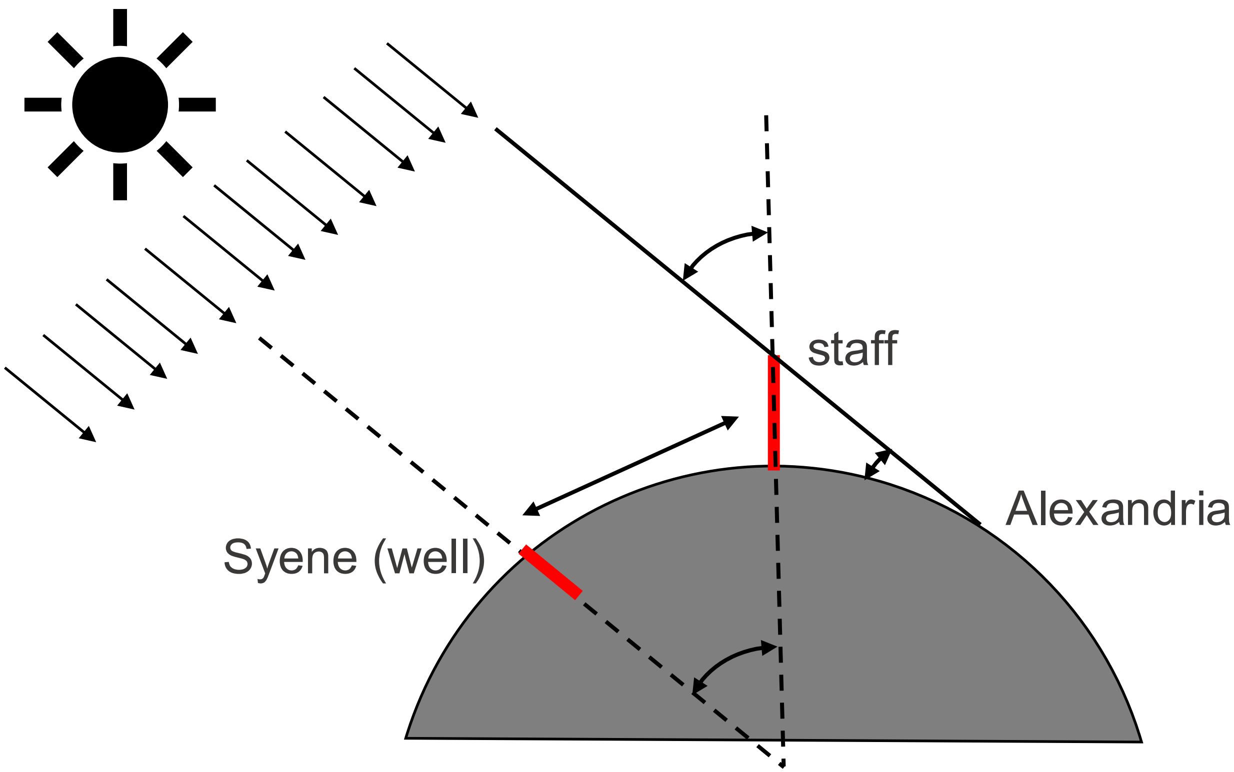
Determining the shape and dimensions of the Earth

- **Aristotle (4th Century BC)**
  - Spherical earth based on gravity
    - All heavy bodies tend to fall towards the center and seek the lowest place in this process. Thus, they are eventually pressed into spherical ball.
  - Observational experiences
    - Ships at sea, the mast is last seen going away from land.
    - Earth cast round shadow during lunar eclipses.

Role of Surveying

Determining the shape and dimensions of the Earth

- ❑ **Eratosthenes (276-195 BC)**
  - ➢ First to make measurement of size of earth.



Role of Surveying

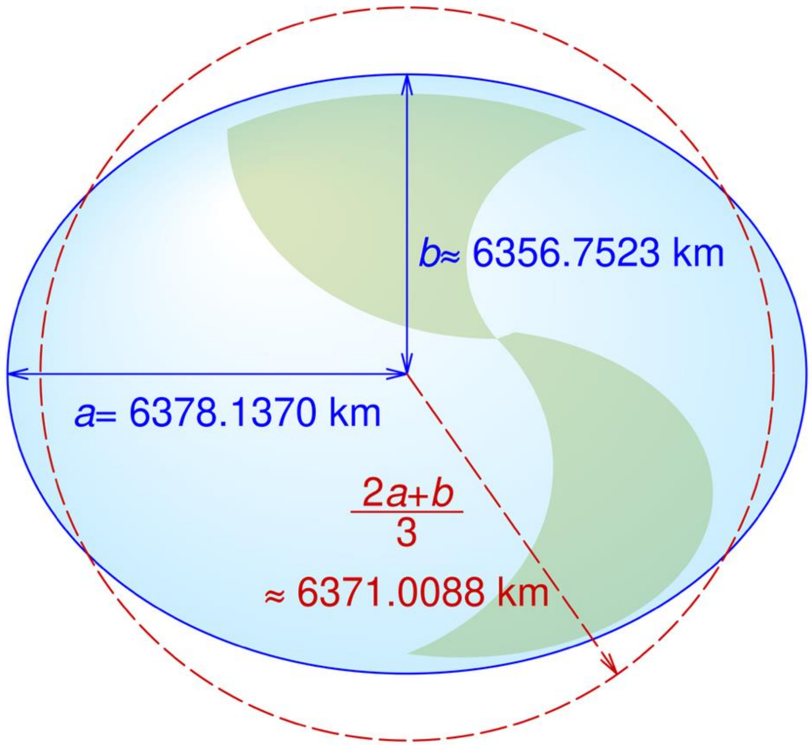
Determining the shape and dimensions of the Earth

- ❑ **Eratosthenes (276-195 BC)**
  - ➢ Errors on Eratosthenes' approach
    - ✓ Sun could not be directly overhead at time of measurement
    - ✓ Alexandria and Syene not on the same meridian
    - ✓ Unit conversion problems

Role of Surveying

Determining the shape and dimensions of the Earth

- ❑ **Shape of the Earth**



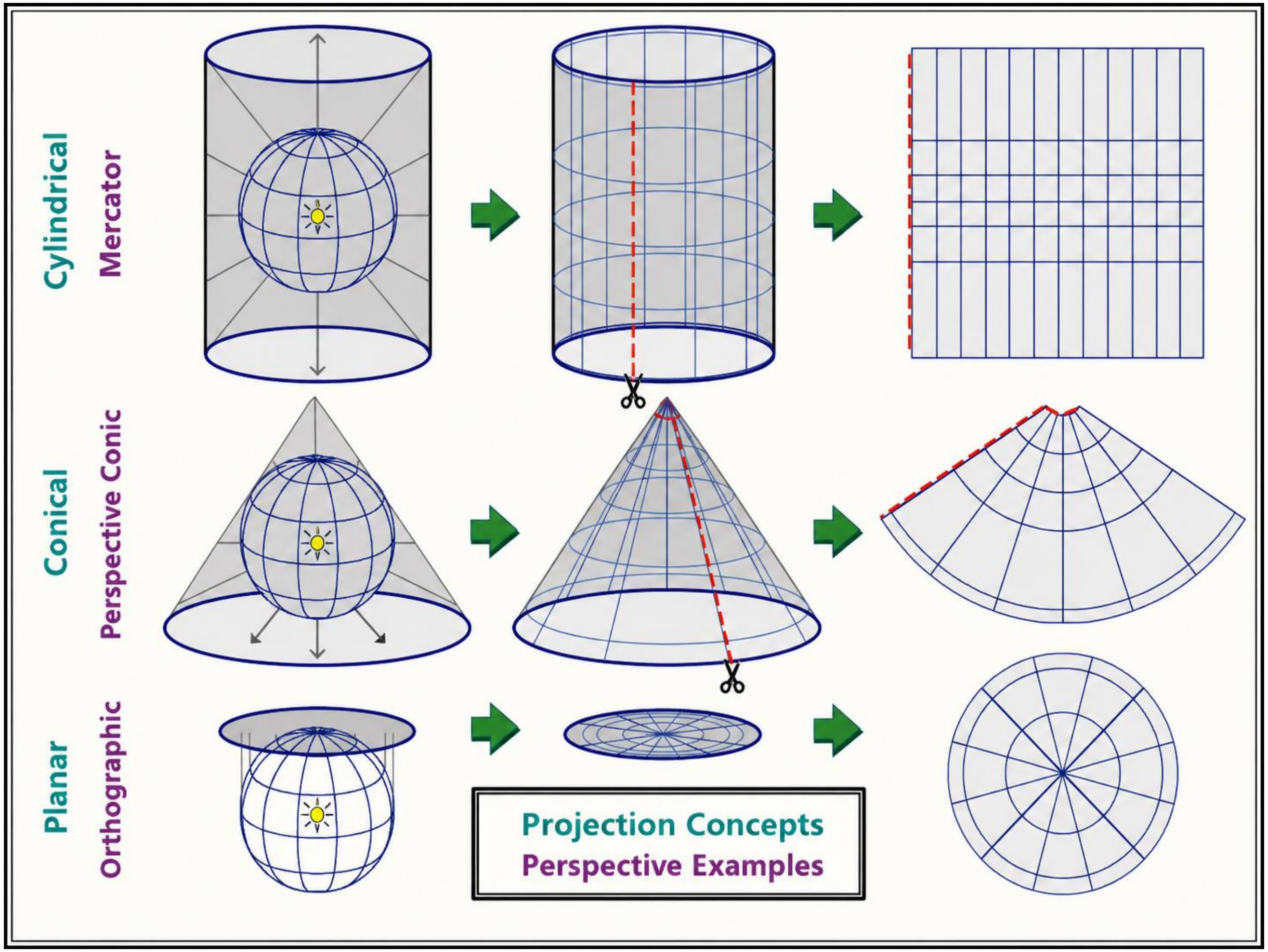
Polar radius = 6356752.3142 meters

Equatorial radius = 6378137.0 meters

Flattening = f = $(a-b)/a$ = 1/298.257

Role of Surveying

Establishing and refining spatial reference frameworks



Role of Surveying

Establishing and refining spatial reference frameworks

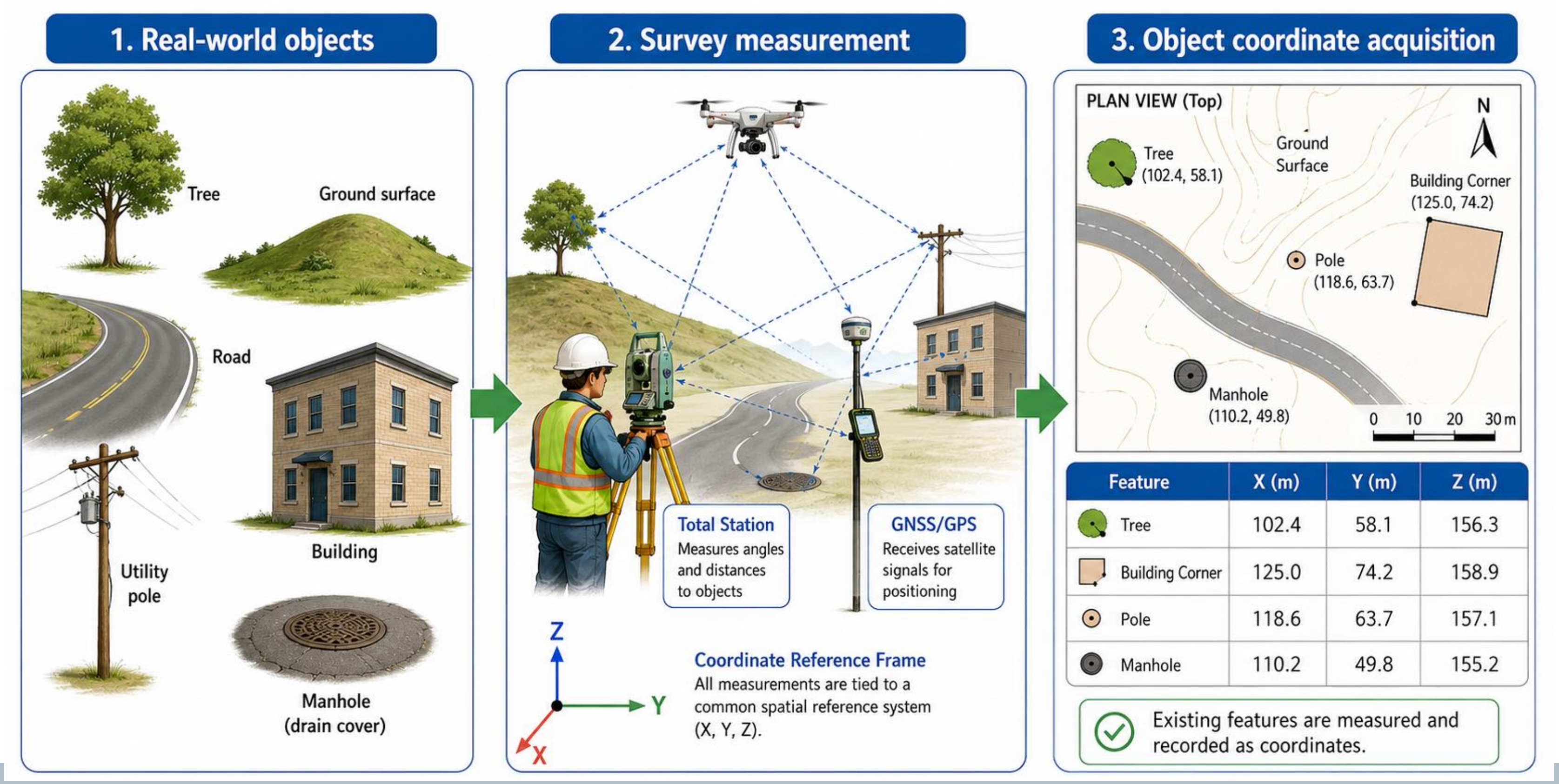
- ❑ **Projections**
  - ➢ Cylindrical
    - ✓ A map projection where earth's surface is projected onto a tangent or secant cylinder, which is then cut lengthwise and laid flat
  - ➢ Conic
    - ✓ A map projection where the earth's surface is projected onto a tangent or secant cone, which is then cut from apex to base and laid flat
  - ➢ Planar
    - ✓ A map projection resulting from the conceptual projection of the earth onto a tangent or secant plane

Role of Surveying

Locating existing natural and constructed features

❑ Locating object

- ➢ Measuring the coordinates of existing natural and constructed features.

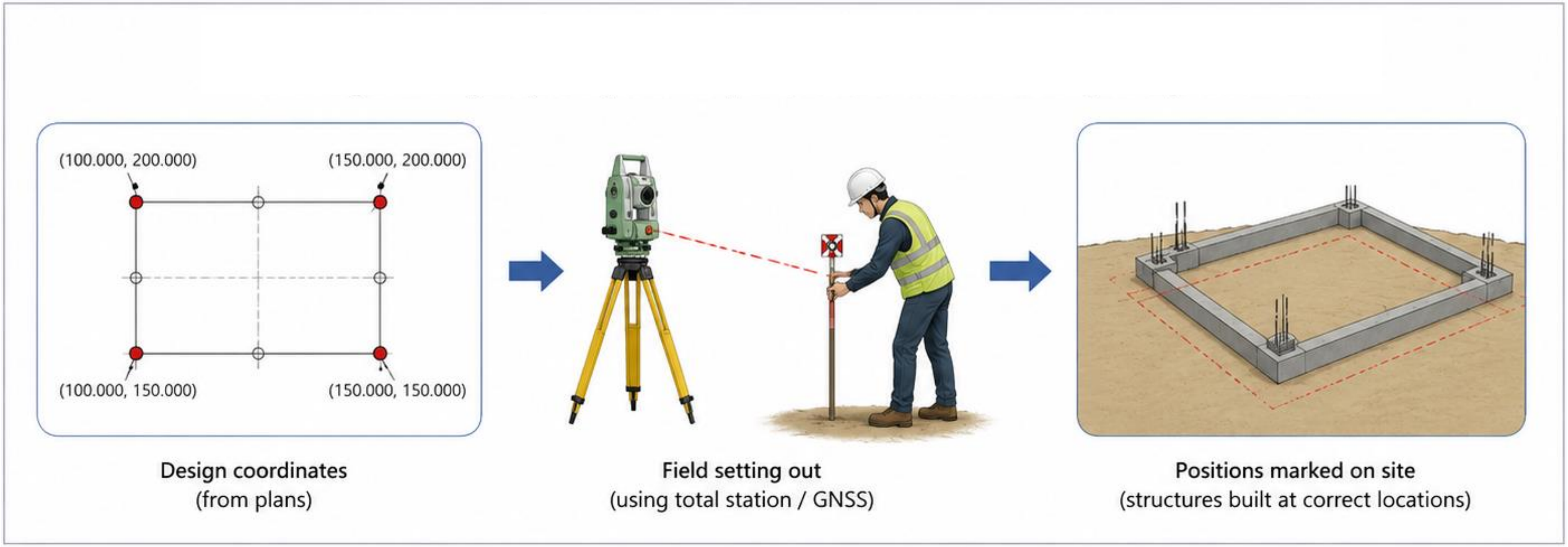


Role of Surveying

Setting out designed positions of engineering structures

❑ Positioning objects

- ➢ Establishing or setting out the precise horizontal and vertical positions of points, structures, or facilities according to design coordinates.



Role of Surveying

Producing maps, plans, charts, and digital survey records

❑ Provision of plans, maps, files, and charts recording surveying products

➢ Producing plans, maps, charts, and digital records that document and communicate survey results.